

Connecting a Backend & Database

Sending Http Requests

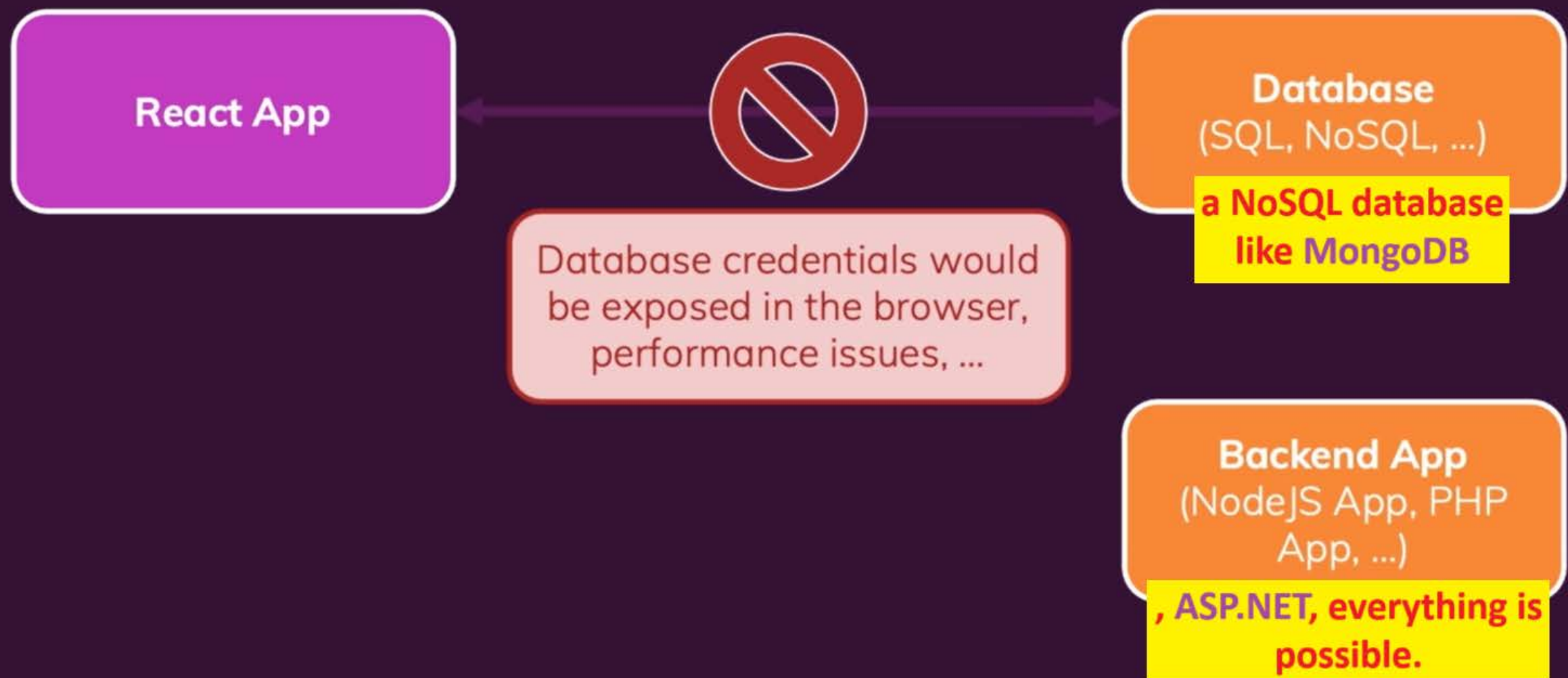
Module Content

How do React Apps Interact with Databases?

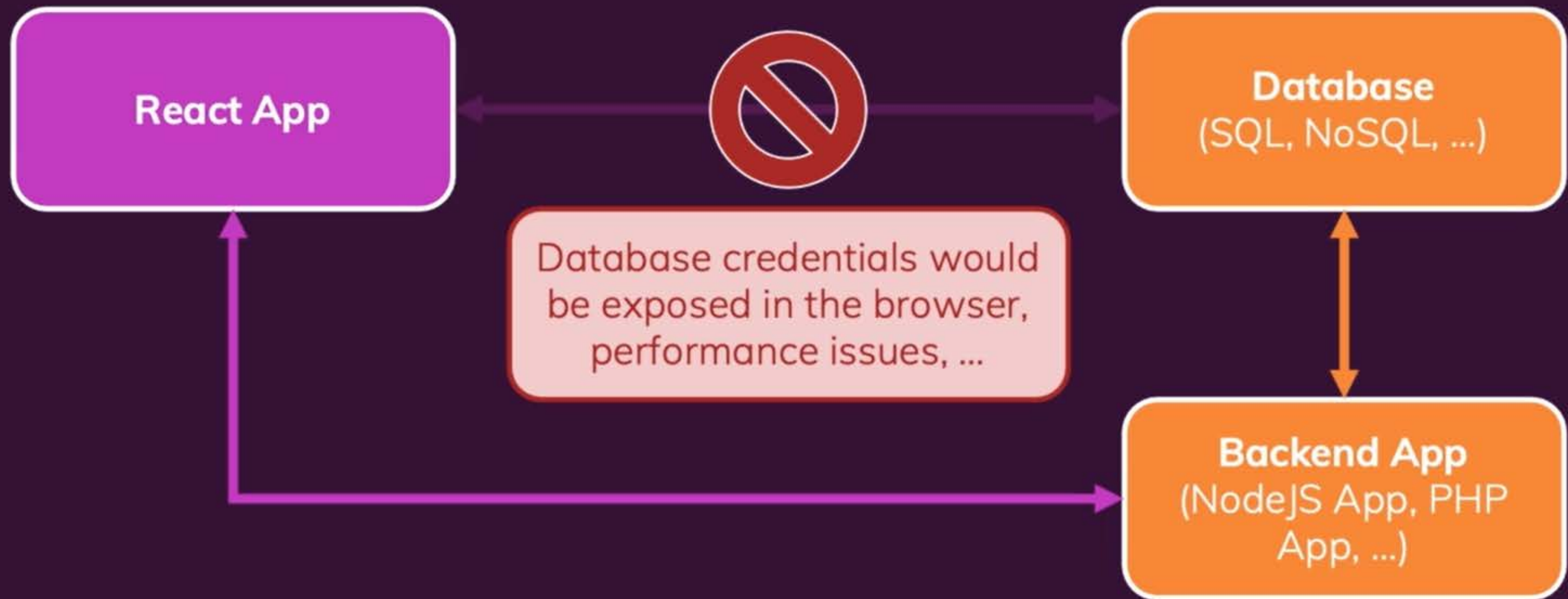
Sending Http Requests & Using Responses

Handling Errors & Loading State

Browser-side Apps Don't Directly Talk To Databases



Browser-side Apps Don't Directly Talk To Databases



SWAPI

The Star Wars API

(what happened to swapi.co?)

All the Star Wars data you've ever wanted:

Planets, Spaceships, Vehicles, People, Films and Species

From all **SEVEN** Star Wars films

Now with The Force Awakens data!

Try it now!

`https://swapi.dev/api/people/1/` request

Need a hint? try *people/1/* or *planets/3/* or *starships/9/*

Result:

```
{
  "name": "Luke Skywalker",
  "height": "172",
  "mass": "77",
  "hair_color": "blond",
  "skin_color": "fair",
  "eye_color": "blue",
  "birth_year": "19BBY",
  "gender": "male",
  "homeworld": "https://swapi.dev/api/planets/1/",
  "films": [
    "https://swapi.dev/api/films/2/",
```

All the Star Wars data you've ever wanted:
Planets, Spaceships, Vehicles, People, Films and Species
From all **SEVEN** Star Wars films
Now with The Force Awakens data!

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https://swapi.dev/api/people/1/

request

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  "homeworld": "https://swapi.dev/api/planets/1/",
  "films": [
    "https://swapi.dev/api/films/2/",
    "https://swapi.dev/api/films/6/",
    "https://swapi.dev/api/films/3/",
    "https://swapi.dev/api/films/1/",
    "https://swapi.dev/api/films/7/"
  ],
  "species": [
```




Js App.js src M X

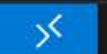


react-the-complete-guide > src > Js App.js > App > fetchMoviesHandler



```
6  function App() {
101
    Complexity is 4.1 Everything is cool!
102  async function fetchMoviesHandler() {
103      setIsLoading(true);
104      setError(null);
105      try {
106          const response = await fetch("https://swapi.dev/api/film/");
107          const data = await response.json();
108
109          if (!response.ok) {
110              // Teh response also has a "status" field which holds the concrete
111              // response status code. You could also manually check that.
112              throw new Error("Something went wrong!");
113          }
114
115          const transformedMovies = data.results.map((movieData) => {
116              return {
117                  id: movieData.episode_id,
118                  title: movieData.title,
119                  openingText: movieData.opening crawl.
```

You, 11 seconds ago • Unco



Fetch Movies

Found no movies.

Unexpected token '<', " <!DOCTYPE "... is not valid JSON

Fetch Movies

Found no movies.

Fetch Movies

Loading...

Fetch Movies

A New Hope

It is a period of civil war. Rebel spaceships, striking from a hidden base, have won their first victory against the evil Galactic Empire. During the battle, Rebel spies managed to steal secret plans to the Empire's ultimate weapon, the DEATH STAR, an armored space station with enough power to destroy an entire planet. Pursued by the Empire's sinister agents, Princess Leia races home aboard her starship, custodian of the stolen plans that can save her people and restore freedom to the galaxy....

The Empire Strikes Back

It is a dark time for the Rebellion. Although the Death Star has been destroyed, Imperial troops have driven the Rebel forces from their hidden base and pursued them across the galaxy. Evading the dreaded Imperial Starfleet, a group of freedom fighters led by Luke Skywalker

✓ REACT-THE-COMPLETE-G...

> .vscode

> public

✓ src

✓ components

 Movie.js

 Movie.module.c...

 MoviesList.js

 MoviesList.mod...

 App.css

 App.js M

 index.css

 index.js

 .gitignore

 package-lock.json

src > App.js > App > fetchMoviesHandler > [🔗] response

```
7  const [movies, setMovies] = useState([]);
8  const [isLoading, setIsLoading] = useState(false);
9  const [error, setError] = useState(null);
10
11  async function fetchMoviesHandler() {
12    setIsLoading(true);
13    setError(null);
14    try {
15      const response = await fetch('https://swapi.dev/api/films/');
16      if (!response.ok) {
17        throw new Error('Something went wrong!');
18      }
19
20      const data = await response.json();
21
22      const transformedMovies = data.results.map((movieData) => {
```

To create a production build, use `npm run build`.



Fetch Movies

Something went wrong!

Google is committed to advancing racial equity for Black communities. [See how.](#)

Firestore helps mobile and web app teams succeed

Get started

Watch the video

Build apps fast, without managing infrastructure

Firestore provides the functionality like analytics, crash reporting so you can focus on your users.

Backed by Google, trusted by top apps

Firestore is built on Google infrastructure and scales automatically, for even the largest apps.

One platform, with products that work better together

Firestore products work great individually but share data and insights, so they work even better together.

This site uses cookies from Google to deliver its services and to analyze traffic.

[More details](#) [OK](#)

Products	Free Spark Plan Generous limits to get started	Pay as you go Blaze Plan Calculate pricing for apps at scale ✓ Free usage from Spark plan included*
A/B Testing		Free
Analytics		Free
App Distribution		Free
App Indexing		Free
Authentication		
Phone Auth - US, Canada, and India ?	10k/month	\$0.01/verification
Phone Auth - All other countries ?	10k/month	\$0.06/verification
Other Authentication services	✓	✓



Incognito (2)

Update



Search

English



Go to console



Sign in

unities. See how.



Anmeldung

Mit Google-Konto anmelden.

E-Mail oder Telefonnummer

[E-Mail-Adresse vergessen?](#)

Nicht ihr Computer? Dann können Sie den Gastmodus verwenden, um sich privat anzumelden.
[Weitere Informationen](#)

[Konto erstellen](#)Weiter



Your Firebase projects


Add project

testing-vue
testing-vue-5ac09
 academind.com

ionic-react-demo
ionic-react-demo-66bdc
 academind.com

flutter-push-demo
flutter-push-demo-51e42
 iOS  academind.com

demo-testing
fir-testing-26082
 </>  academind.com

Explore a demo project

 iOS

Firebase projects are containers for your apps

Apps in a project share features like Realtime Database and Analytics

 [Learn more](#)

✕ Create a project(Step 1 of 3)

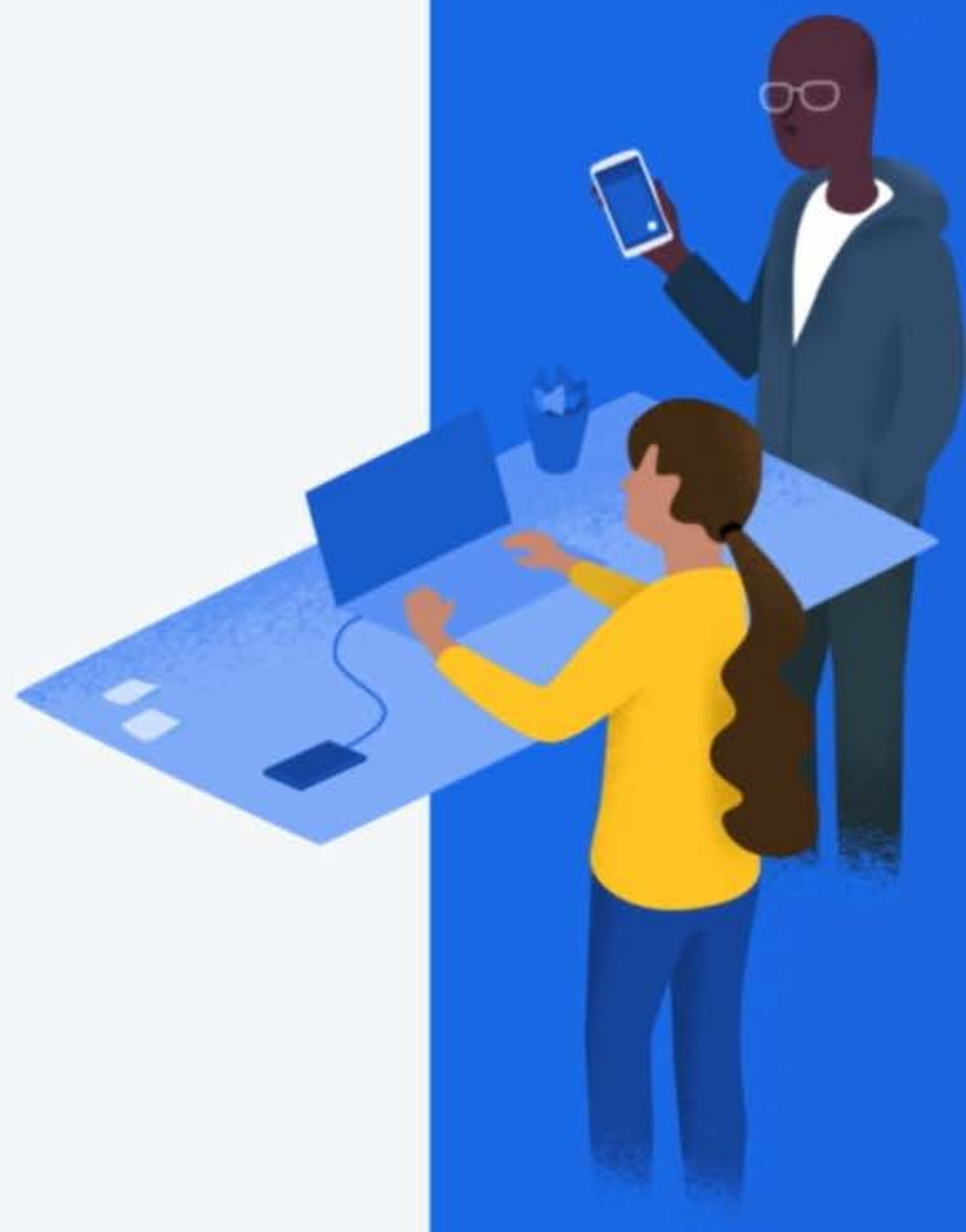
Let's start with a name for your project?

Project name

react-http

- react-http-6b4a6
- academind.com

Continue



×

Create a project(Step 2 of 2)

Google Analytics

for your Firebase project

Google Analytics is a free and unlimited analytics solution that enables targeting, reporting and more in Firebase Crashlytics, Cloud Messaging, In-App Messaging, Remote Config, A/B Testing, Predictions and Cloud Functions.

Google Analytics enables:

- ×
- A/B testing ⓘ
- ×
- Crash-free users ⓘ
- ×
- User segmentation and targeting across Firebase products ⓘ
- ×
- Event-based Cloud Functions triggers ⓘ
- ×
- Predicting user behaviour ⓘ
- ×
- Free unlimited reporting ⓘ

☐

Enable Google Analytics for this project

👤recommended

[Previous](#)

Create project





Finishing up...
react-http



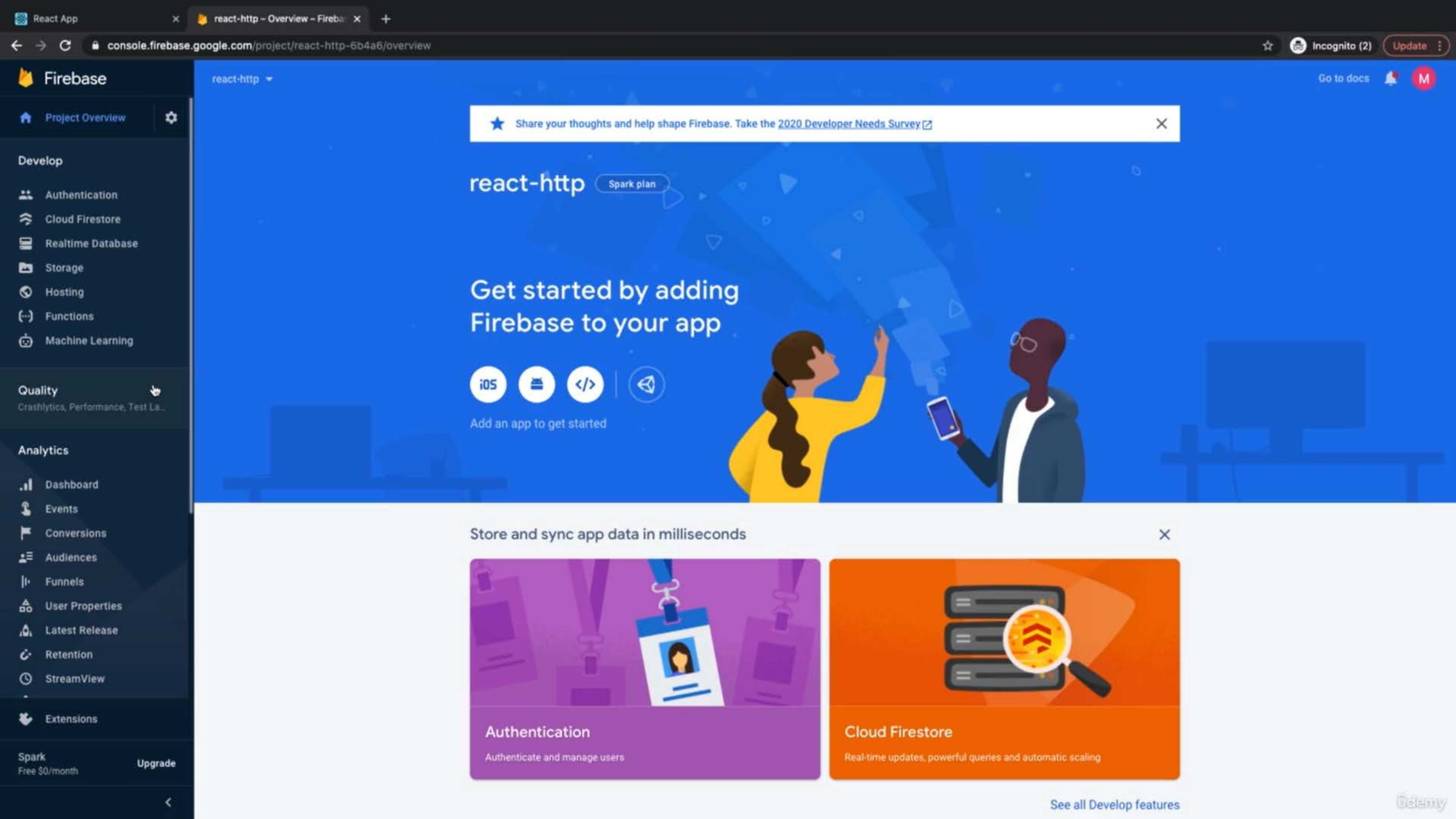


react-http

✔ Your new project is ready

Continue







Project Overview



Develop

Authentication

Cloud Firestore

Realtime Database

Storage

Hosting

Functions

Machine Learning

Quality

Crashlytics, Performance, Test La...

Analytics

Dashboard

react-http ▾



Share your thoughts and help shape Firebase. Ta

react-http

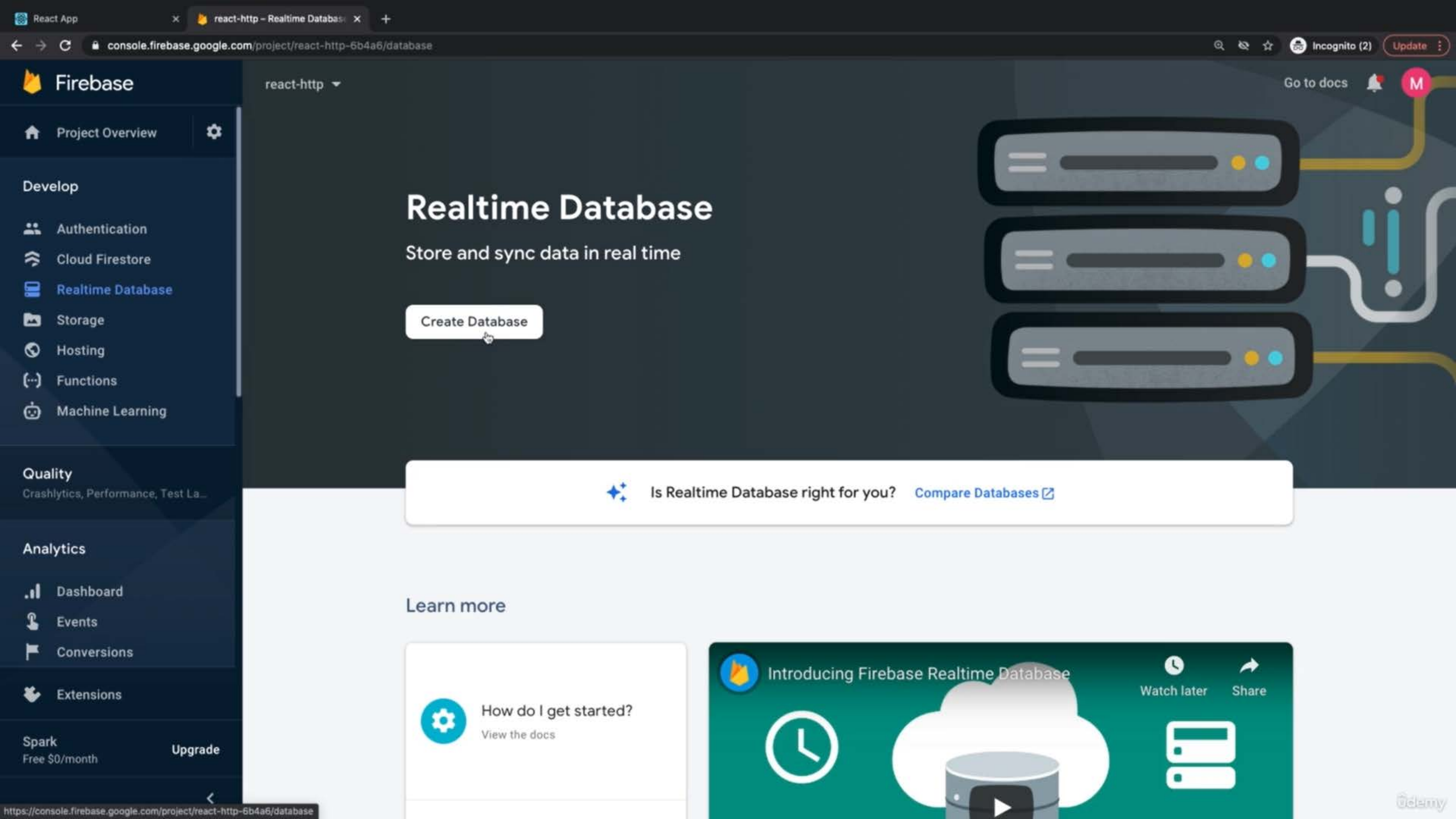
Spark plan

Get started by adding Firebase to your app

iOS



Add an app to get started



Realtime Database

Store and sync data in real time

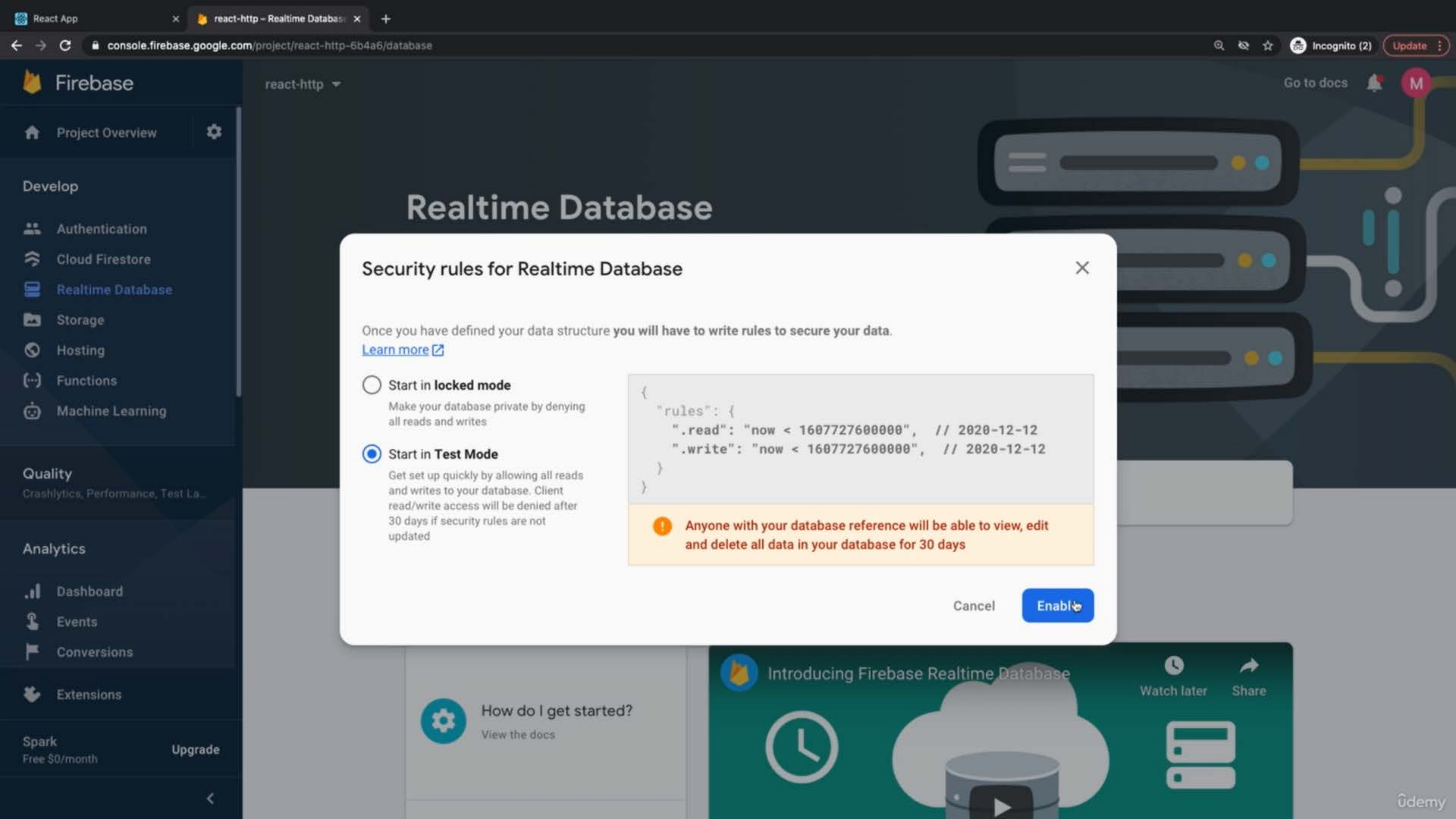
Create Database

Is Realtime Database right for you? [Compare Databases](#)

Learn more

How do I get started?
[View the docs](#)





- Project Overview
- Develop
 - Authentication
 - Cloud Firestore
 - Realtime Database
 - Storage
 - Hosting
 - Functions
 - Machine Learning
- Quality
 - Crashlytics, Performance, Test Lab
- Analytics
 - Dashboard
 - Events
 - Conversions
 - Extensions
- Spark
 - Free \$0/month
- Upgrade

Realtime Database

Security rules for Realtime Database

Once you have defined your data structure you will have to write rules to secure your data.
[Learn more](#)

- ☐ Start in **locked mode**
Make your database private by denying all reads and writes
- ☒ Start in **Test Mode**
Get set up quickly by allowing all reads and writes to your database. Client read/write access will be denied after 30 days if security rules are not updated

```
{
  "rules": {
    ".read": "now < 1607727600000", // 2020-12-12
    ".write": "now < 1607727600000", // 2020-12-12
  }
}
```

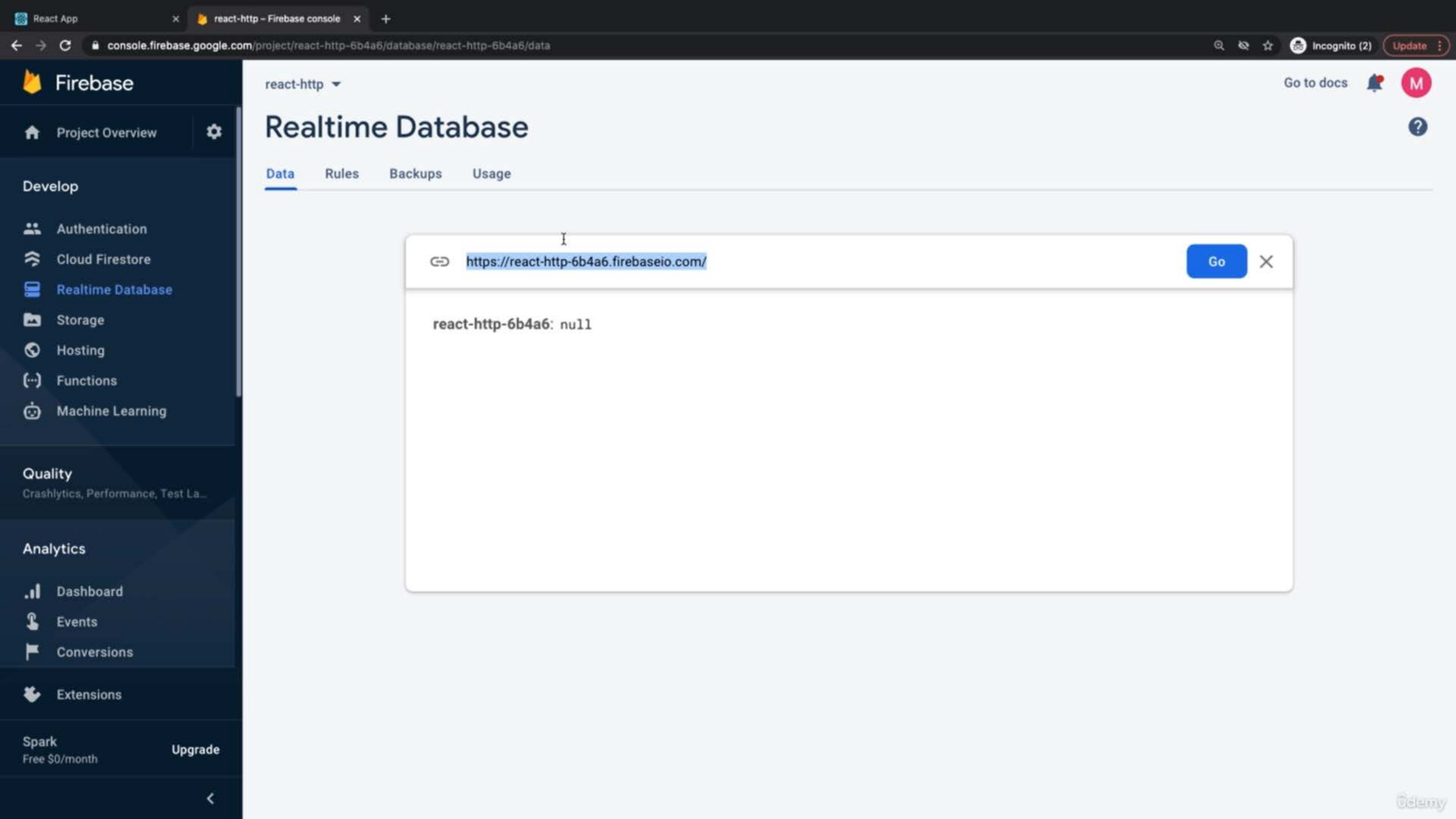
⚠ Anyone with your database reference will be able to view, edit and delete all data in your database for 30 days

Cancel Enable

How do I get started?
[View the docs](#)

Introducing Firebase Realtime Database

Watch later Share



EXPLORER

JS App.js



JS AddMovie.js

AddMovie.module.css



REACT-THE-COMPLETE-G...

> .vscode

> public

src

components

JS AddMovie.js

AddMovie.mod...

JS Movie.js

Movie.module.c...

JS MoviesList.js

MoviesList.mod...

App.css

JS App.js

index.css

JS index.js

src > JS App.js > App > fetchMoviesHandler > useCallback() callback > response

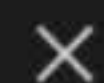
```
11
12  const fetchMoviesHandler = useCallback(async () => {
13    setIsLoading(true);
14    setError(null);
15    try {
16      const response = await fetch('https://swapi.dev/api/films/');
17      if (!response.ok) {
18        throw new Error('Something went wrong!');
19      }
20
21      const data = await response.json();
22
23      const transformedMovies = data.results.map((movieData) => {
24        return {
25          id: movieData.episode_id,
```

TERMINAL

PROBLEMS



1: node



To create a production build, use `npm run build`.



EXPLORER

...

JS App.js

●

JS AddMovie.js

AddMovie.module.css



...

✓ REACT-THE-COMPLETE-G...

> .vscode

> public

✓ src

✓ components

JS AddMovie.js

AddMovie.mod...

JS Movie.js

Movie.module.c...

JS MoviesList.js

MoviesList.mod...

App.css

JS App.js

index.css

JS index.js

src > JS App.js > App > fetchMoviesHandler > useCallback() callback > response

```
10   [!], selector] = useState(null);
11
12   chMoviesHandler = useCallback(async () => {
13     ading(true);
14     r(null);
15
16     response = await fetch('https://react-http-6b4a6.firebaseio.com/');
17     response.ok) {
18       ow new Error('Something went wrong!');
19
20
21     data = await response.json();
22
23     transformedMovies = data.results.map((movieData) => {
24       urn {
25         d: movieData.episode_id,
```

TERMINAL

PROBLEMS

...

1: node

v

+

□

🗑

^

✕

To create a production build, use `npm run build`.



REACT-THE-COMPLETE-G...

> .vscode

> public

src

components

JS AddMovie.js

AddMovie.mod...

JS Movie.js

Movie.module.c...

JS MoviesList.js

MoviesList.mod...

App.css

JS App.js

index.css

JS index.js

src > JS App.js > App > fetchMoviesHandler > useCallback() callback > response

10 `setError() = useState(null);`

11

12 `esHandler = useCallback(async () => {`13 `true);`14 `};`

15

16 `nse = await fetch('https://react-http-6b4a6.firebaseio.com/movies.json');`17 `se.ok) {`18 `Error('Something went wrong!');`

19

20

21 `= await response.json();`

22

23 `formedMovies = data.results.map((movieData) => {`

24

25 `ieData.episode_id,`

* So this name is up to you, and I choose movies because I want to store some movies.

* A (.json) is important here. That's something Firebase specific, they need this (.json) at the end of the URL you are sending requests to, otherwise your requests will fail.

TERMINAL

PROBLEMS

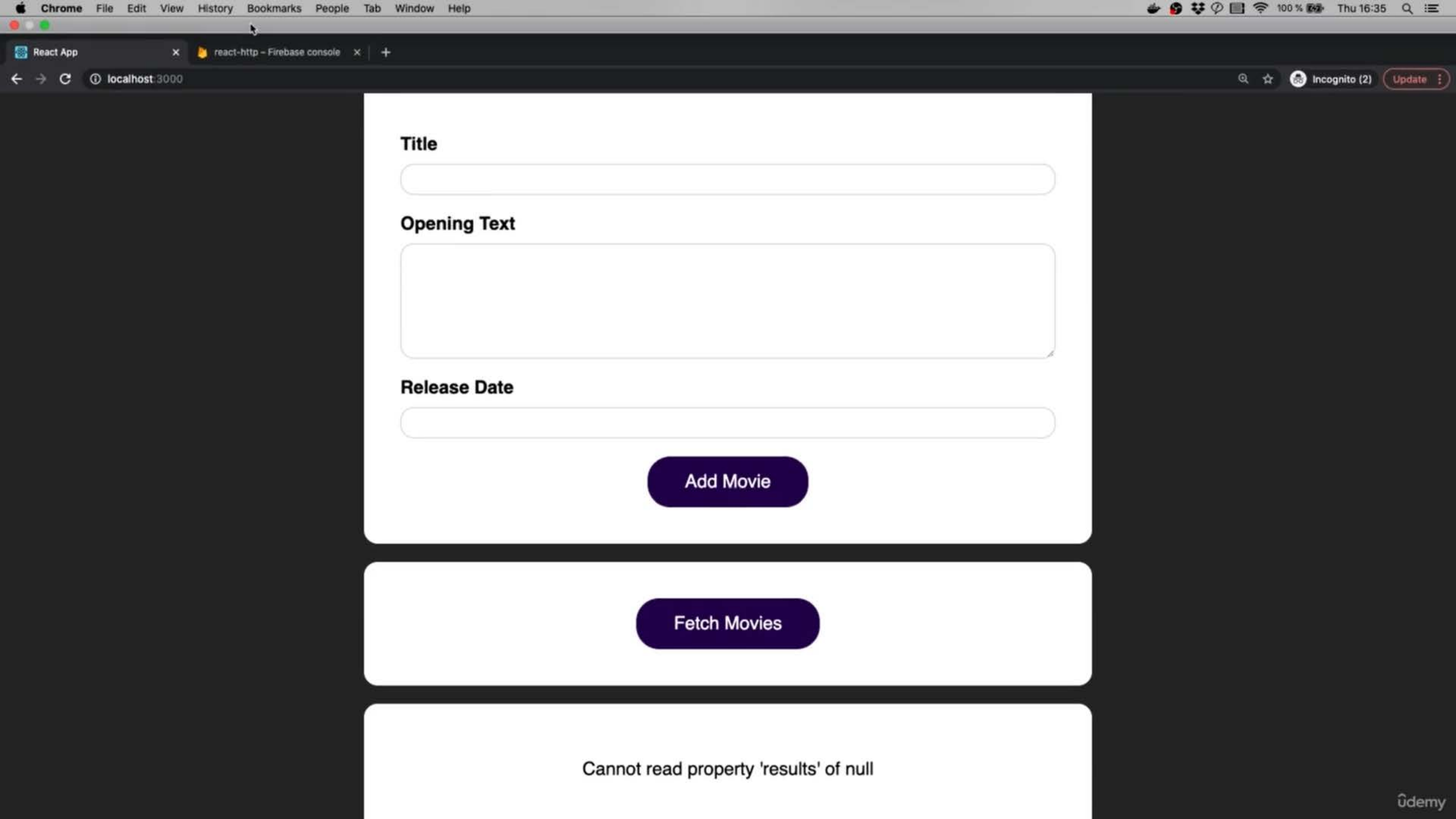
...

1: node

v

+

To create a production build, use `npm run build`.



Title

Opening Text

Release Date

Add Movie

Fetch Movies

Cannot read property 'results' of null

Title

My first movie

Opening Text

Some text

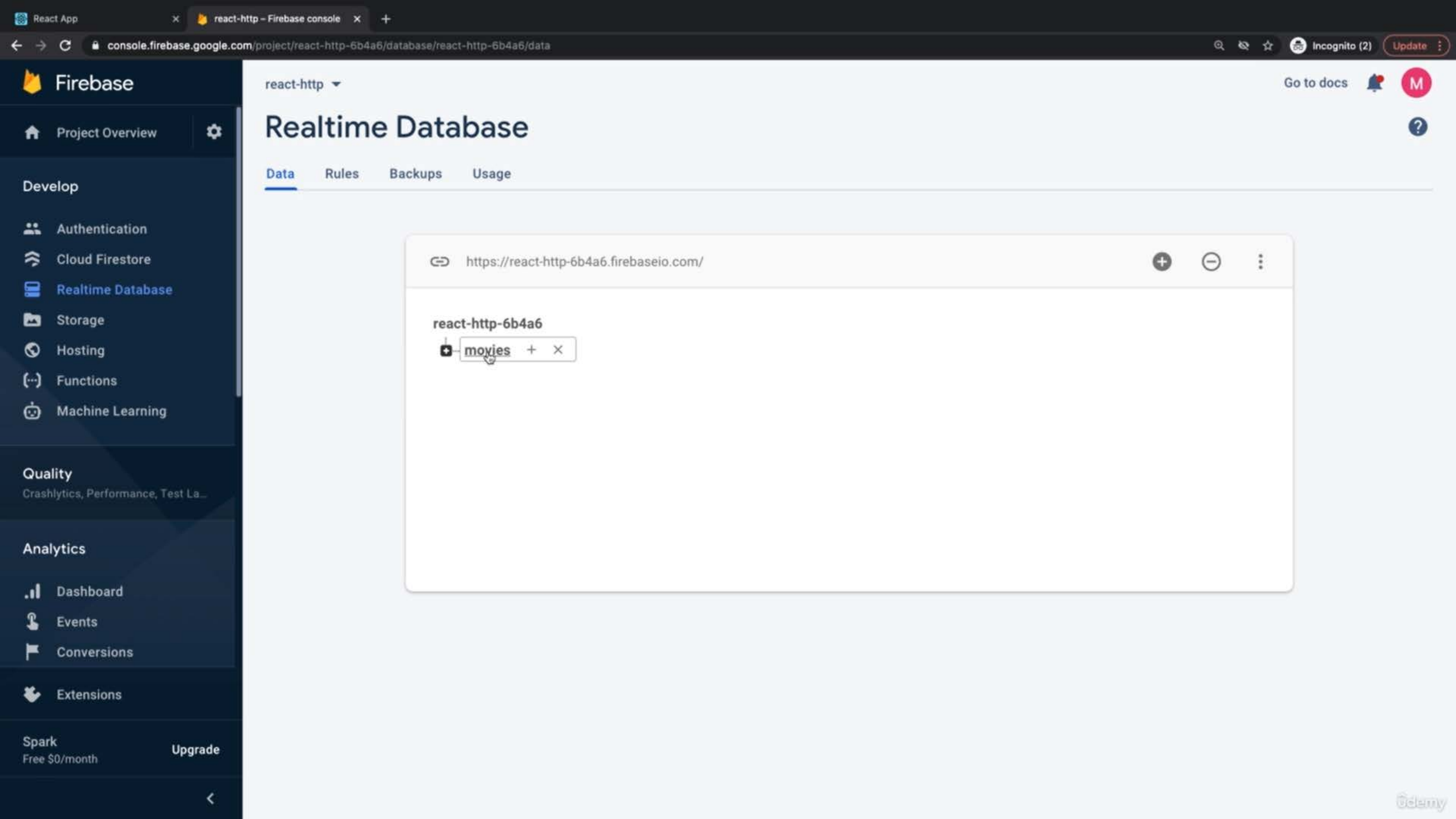
Release Date

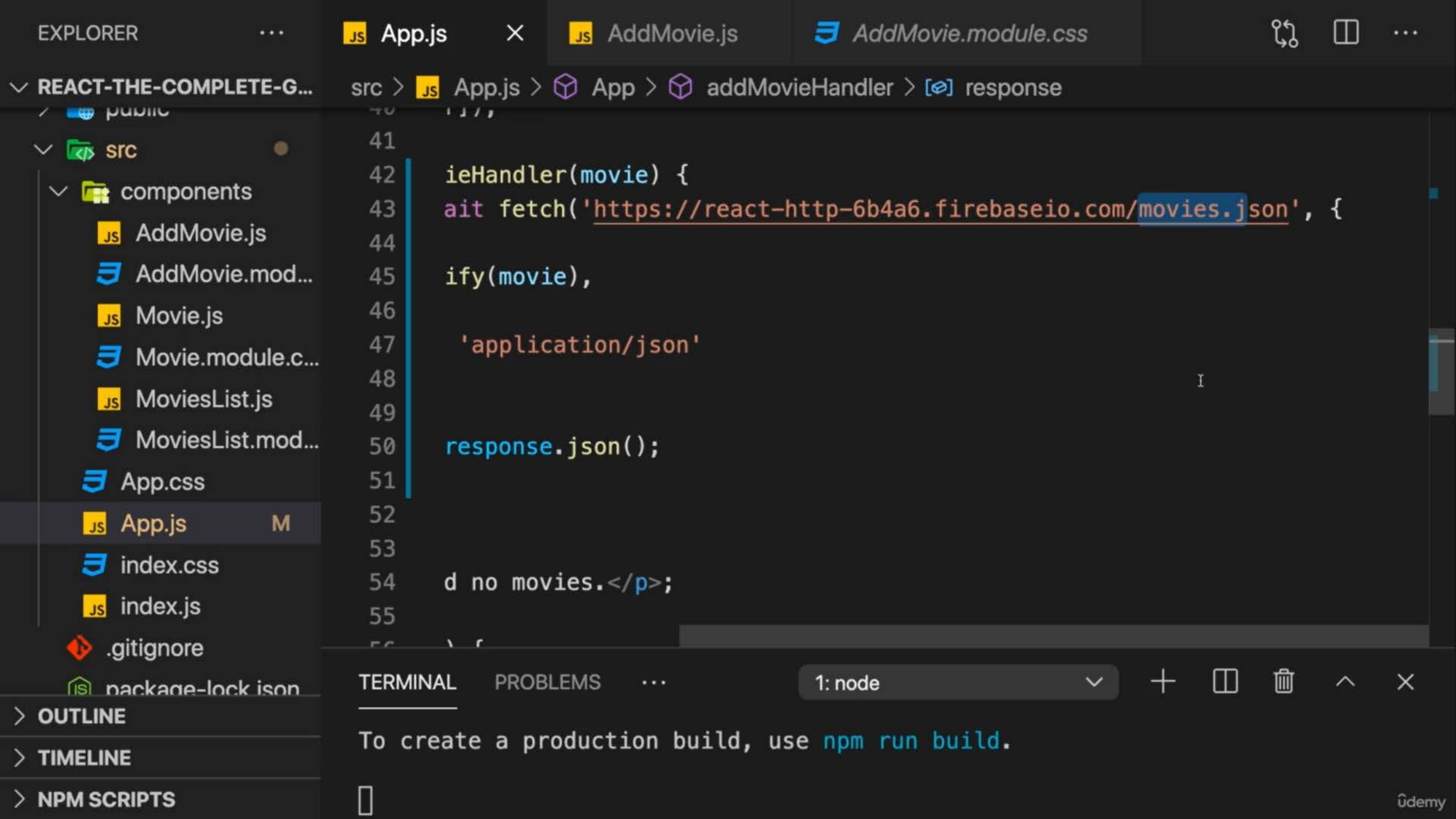
2020-06-01

Add Movie

Fetch Movies

Cannot read property 'results' of null





EXPLORER

App.js

AddMovie.js

AddMovie.module.css

REACT-THE-COMPLETE-G...

public

src

components

AddMovie.js

AddMovie.mod...

Movie.js

Movie.module.c...

MoviesList.js

MoviesList.mod...

App.css

App.js

M

index.css

index.js

.gitignore

package-lock ison

OUTLINE

TIMELINE

NPM SCRIPTS

src > App.js > App > addMovieHandler > response

```
41  
42   ieHandler(movie) {  
43     ait fetch('https://react-http-6b4a6.firebaseio.com/movies.json', {  
44  
45     ify(movie),  
46  
47     'application/json'  
48  
49  
50     response.json();  
51  
52  
53  
54     d no movies.</p>;  
55
```

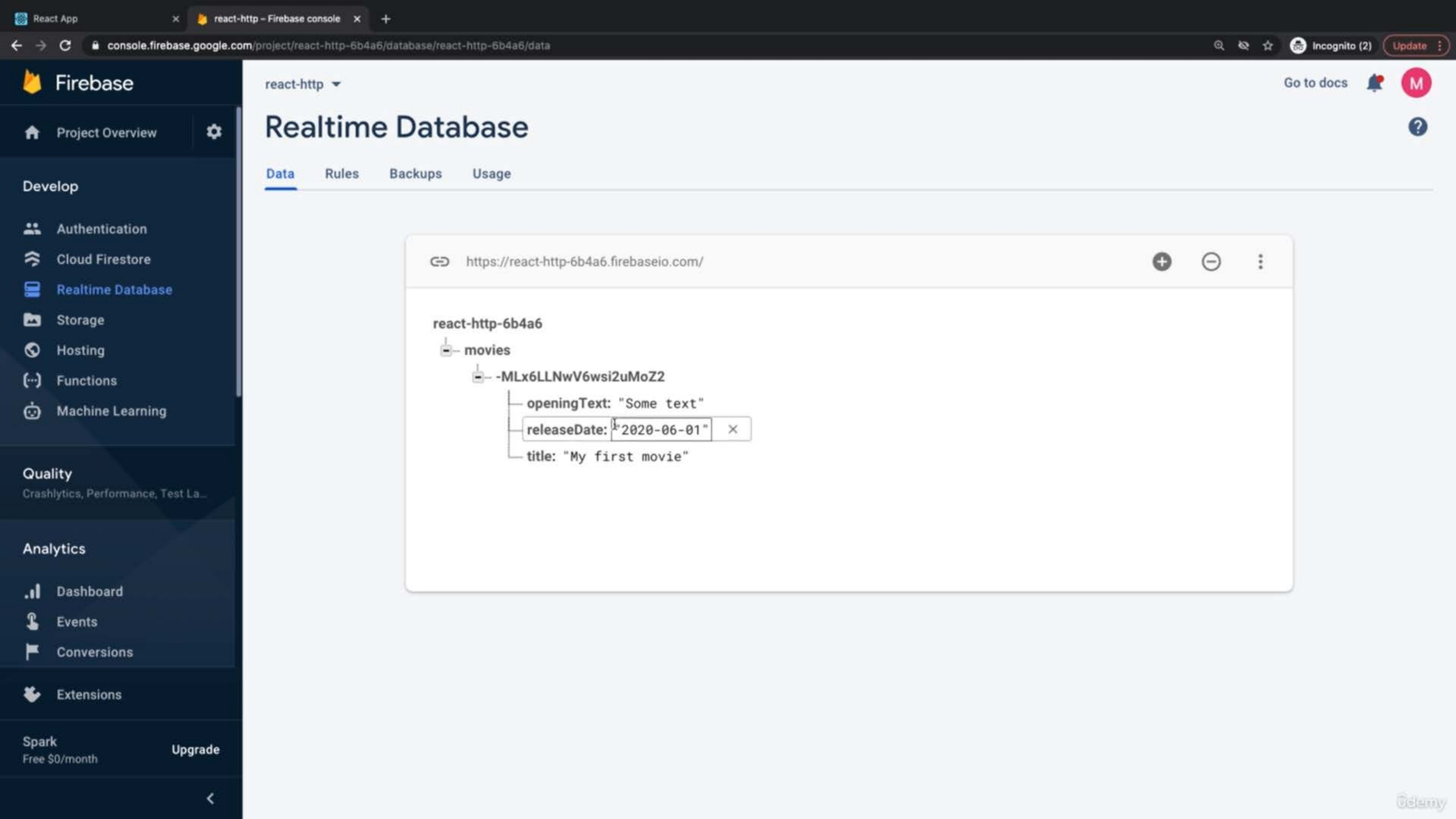
TERMINAL

PROBLEMS

1: node

To create a production build, use `npm run build`.

Udemy



React App

react-http - Firebase console

localhost:3000

Title

My first movie

Opening Text

Some text

Release Date

2020-06-01

Add Movie

Fetch Movies

Cannot read property 'results' of null

Incognito (2) Update

Console

top

Filter

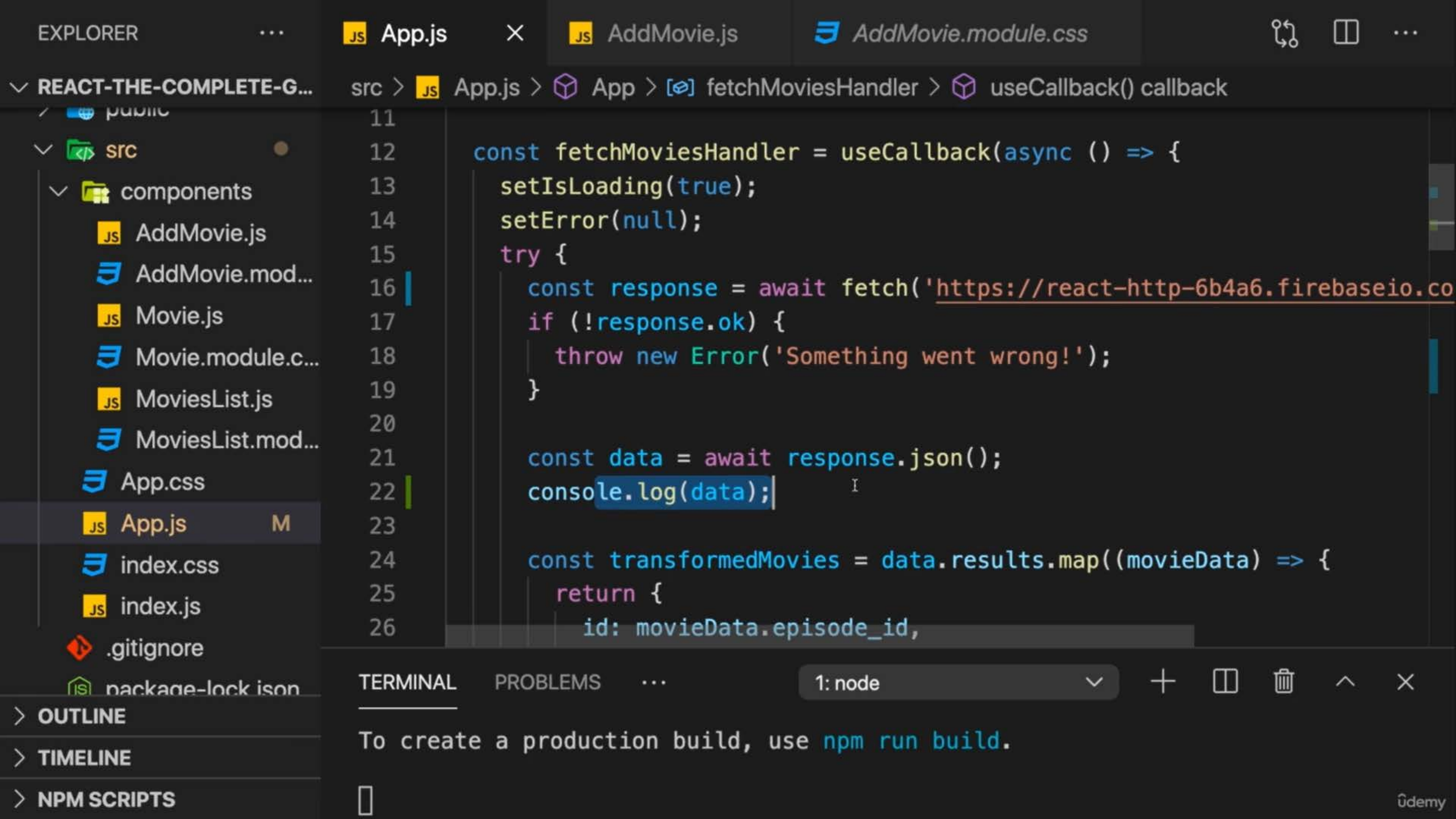
[HMR] Waiting for update signal from WDS... log.js:24

react-dom.development.js:25028
Download the React DevTools for a better development experience: <https://fb.me/react-devtools>

App.js:51

▼ Object *i*
name: "-MLx6LLNwV6wsi2uMoZ2"
▶ __proto__: Object

>



React App

react-http - Firebase console

localhost:3000

Title

Opening Text

Release Date

Add Movie

Fetch Movies

Cannot read property 'map' of undefined

Incognito (2) Update

Console

top

Filter

[HMR] Waiting for update signal from WDS... log.js:24

react-dom.development.js:25028
Download the React DevTools for a better development experience: <https://fb.me/react-devtools>

App.js:22

▼ {-MLx6LLNwV6wsi2uMoZ2: {...}}

▼ -MLx6LLNwV6wsi2uMoZ2:
openingText: "Some text"
releaseDate: "2020-06-01"
title: "My first movie"
▶ __proto__: Object
▶ __proto__: Object

>

Udemy

React App

react-http - Firebase console

localhost:3000

Title

A Second Movie

Opening Text

With more text!

Release Date

2022-08-21

Add Movie

Fetch Movies

Movie

Incognito (2) Update

Console

top

Filter

[HMR] Waiting for update signal from WDS... log.js:24

react-dom.development.js:25028

Download the React DevTools for a better development experience: <https://fb.me/react-devtools>

>

Odeemy

React App

react-http - Firebase console

localhost:3000

With more text!

Release Date

2022-08-21

Add Movie

Fetch Movies

My first movie

Some text

Incognito (2) Update

Console >>

top

Filter

[HMR] Waiting for update signal from WDS... log.js:24

react-dom.development.js:25028
Download the React DevTools for a better development experience: <https://fb.me/react-devtools>

App.js:54
▶ {name: "-MLx7BM1Y8qVoLrTCuQU"}

>

